Summary of the "Decreasing Universe" Theory

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Introduction

The Decreasing Universe (D.U.) theory proposes that the gravitational field continuously contracts space and everything within it.

This effect provides an alternative explanation for two key cosmological phenomena:

- The apparent accelerated expansion of galaxies, without requiring dark energy.

- The anomalous orbital velocities of stars, without invoking dark matter.

At first glance, this idea may seem counterintuitive, but relativity already provides two analogous effects:

1. Equivalence Principle – In an accelerating elevator, space and objects inside it undergo continuous contraction in the direction of motion.

2. Schwarzschild Metric – Near a massive body, radial distances are altered, causing space to become "compressed."

Although the D.U. theory does not directly derive from these effects, they illustrate that gravitationally induced contraction is not an unreasonable concept.

Dark Energy and the Expansion of the Universe

The D.U. theory explains the apparent accelerated expansion of galaxies as a consequence of local space contraction.

Since we are within a gravitational field, the measuring instruments we use—including atomic clocks and rulers—are also shrinking.

As a result, when we observe distant galaxies, they appear to be moving away faster than expected, even if their actual motion remains unchanged.

This contraction-based interpretation allows us to derive Hubble's Law, offering an alternative to the dark energy hypothesis.

For more details, see the paper:
"Derivation of Hubble’s Law and its relation to dark energy and dark matter."

Dark Matter and Galactic Rotation Curves

The Dark Matter Effect is traditionally used to explain why stars in the outer regions of galaxies move faster than expected based on visible mass alone.

However, the D.U. theory provides an alternative explanation: continuous galactic contraction due to gravity.

- Closer to the galactic center, the contraction rate is higher, leading to a greater reduction in emitted photon wavelengths.

- Farther from the center, the contraction rate decreases, meaning photon wavelengths are less compressed.

Since stellar velocities are measured using redshift (z), this differential contraction mimics the effect of an increased Doppler shift for stars farther from the galactic center.

As a result, their velocities appear higher than expected—without requiring an invisible "dark matter" component.

(I am currently developing a formula to quantify the contraction rate as a function of gravitational intensity.)

Testing the Theory Against the ΛCDM Model

The current standard cosmological model, ΛCDM, postulates Dark Energy and Dark Matter to explain these observed phenomena.

However, the Decreasing Universe theory makes a testable prediction:

- Among galaxies at the same distance from Earth, more massive galaxies should have a lower redshift due to their stronger gravitational contraction.

- The ΛCDM model predicts the opposite, as it assumes a gravitational redshift effect that would increase redshift in more massive galaxies.

To test this, I compiled a dataset comparing mass and redshift.

The data supports the Decreasing Universe model and contradicts the ΛCDM predictions.

For a full analysis, see:

"Decreasing Universe: Redshifts and Distance Data Refute the L-CDM Model."

Conclusion

The Decreasing Universe model provides a unified explanation for both:

- The apparent acceleration of the universe's expansion (without dark energy).

- The anomalous galactic rotation curves (without dark matter).

By reconsidering the role of gravitational contraction, this theory offers a new perspective on cosmology—one that challenges the need for undetected exotic components in the universe.

References

[01] Derivation of Hubble's Law and its relation to dark energy and dark matter
https://medcraveonline.com/OAJMTP/OAJMTP-02-00049.pdf

[02]Decreasing universe: the distance as a function of redshift
https://medcraveonline.com/AAOAJ/decreasing-universe-the-distance-as-a-function-of-redshift.html

[03]Decreasing Universe: Redshifts and Distance Data Refute the L-CDM Model
https://www.wecmelive.com/open-access/decreasing-universe-redshifts-and-distance-data-refute-the-lcdm-model.pdf

[04] What if the Universe Is NOT Expanding? (Douglas Giles)
https://insertphilosophyhere.com/what-if-the-universe-is-not-expanding/